Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/26/2016
PI Name:	O'Brian, Thomas Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chris.oates@nist.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	303-497-4570
Organization Name:	National Institute of Standards and Technology		
PI Address 1:	Time and Frequency Division		
PI Address 2:	325 Broadway Street, MS 847.00		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/01/2012	**End Date:**	11/15/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/15/2015 due to change in Principal Investigator, per PI and new PI Oates (Ed., 12/17/15)		
Key Personnel Changes/Previous PI:	Project closes with Thomas O'Brian as PI due to reduced workload; project continues with new PI Chris Oates, Ph.D. See project with the same title and Oates as the PI for subsequent reporting.		
COI Name (Institution):	Jefferts, Steven (National Institute of Standards & Technology)		
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:
1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following:

• Global time and frequency transfer with uncertainty as low as 10-17.

• Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10-17 or better.

• Test of gravitation redshift at 2 x 10-6 uncertainty or better.

• Search for variation in fundamental constants, such as fine structure constant at level of 10-18 per year or better.

All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list):

• World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10-16 or better.

• Two mercury ion optical clocks with current fractional frequency uncertainty of approximately 1 x 10-17 and continually improving.

• Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 8 x 10-18 and continually improving.

• Two ytterbium lattice optical clocks with uncertainties better than 1 x 10-16 and rapidly improving.

• Two strontium lattice optical clocks with uncertainties approximately 3 x 10-17 and rapidly improving. (At JILA. A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.)

• A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons.

• A world-class time scale comprising seven hydrogen masers, with eight more masers to be added over the next three years.

NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications.

In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment.

We propose the following revised program plan outline:

Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction.

Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure.

Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time.

Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The major benefits and impacts of this research program can be grouped into three main areas:
1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements.

2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from

ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory.

3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.

Task Progress:

In the most recent reporting period (April 2014 through November 14, 2015) work has focused on three main tasks:
1. Preparation of the NIST site for the Microwave Link Ground Terminal (MWL GT).

2. Advancing the NIST time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics.

3. Theory support for ACES goals for precision time and frequency transfer and tests of fundamental physics.

Progress on each task is described below.

1. Preparation of the NIST site for the Microwave Link Ground Terminal (MWL GT).

The Microwave Link Ground Terminal (MWL GT) system will be provided by ESA to participating ground station partners across the world, including NIST in Boulder, Colorado. However, the MWL GT systems are still under development through ESA sponsorship. In the meantime, NIST has been closely coordinating with ESA contacts on details of the MWL GT hardware, information technology, and infrastructure requirements.

During the reporting period NIST has made substantial progress on the construction of the hardware that will host the ground terminal. NIST scientists worked with NIST facilities staff and engineering firms to develop the best location for the MWL GT, one that provides the best available views of the ISS as it passes by in orbit. Together we designed and developed the physical and support infrastructure needed for the MWL GT, the required utilities and information technology support, and other required infrastructure. The optimal site was located on the roof of the new $100 million NIST Precision Measurement Laboratory, as it provides the best view of the ISS, the best infrastructure to support the MWL GT hardware, and a location for easy access to signals from NIST microwave and optical frequency standards. Construction of the platform, as well as its supporting electrical infrastructure is now complete.

2. Advancing the NIST time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics.

NIST continues to perform research on and improve the performance of its several optical and microwave frequency standards that will be crucial to international time and frequency comparisons and ACES-mediated fundamental research: two single-aluminum ion optical standards, one single-mercury ion optical standard, two ytterbium lattice optical standards, two strontium lattice optical standards, two cesium microwave primary standards, and multiple other frequency standards under development including calcium beam optical standards. These standards are among the best in the world and as such will be crucial players in the world clock comparisons enabled by ACES.

The past project year included substantial advances in several of NIST's frequency standards:

• The NIST-F2 cesium fountain primary frequency standard submitted its first formal report to BIPM (International Bureau of Weights and Measures) on the "Duration of the TAI scale interval," essentially the frequency of the SI second. NIST-F2 has a Type B uncertainty of 1.1 x 10-16, the smallest uncertainty of any of the approximately one dozen primary frequency standards currently in operation across the world. NIST-F1 will continue to operate as well, with a current Type B uncertainty of 3.1 x 10-16. The two high-performance NIST primary frequency standards will support general ACES goals of precision international time and frequency transfer, inter-comparison of frequency standards across the world, and fundamental physics experiments. Full details on the NIST-F2 frequency evaluation has been published in issue of Metrologia (see Publications section).

• The NIST ytterbium and the NIST/JILA strontium lattice optical clocks have reduced their fractional frequency uncertainty to below 5 x 10-18, a level they can now reach with only about fifteen minutes of averaging. This ultimate stability enables the lattice clocks to be exceptionally precise standards for fundamental physics research. This remarkable stability dramatically advances the goals of ACES time and frequency transfer given the relatively short visibility of the ISS to any MWL GT.

3. Theory support for ACES goals for precision time and frequency transfer and tests of fundamental physics.

A critical aspect of the ACES mission is accurate transfer of the time and frequency data between the ISS and the ground stations. In particular, precise determination of the ISS orbit is essential to optimize the time/frequency transfer accuracy. Thus, during the reporting period, NIST's efforts in support of the ACES goals have principally been focused on the best methods for orbit determination through data analysis techniques. What makes this challenge particularly complex is the fact that data from the ACES equipment on the ISS will come not only from GPS receivers that estimate position and velocity, but from two-way ranging between ground-based monitors and the ISS, and possibly Doppler measurements of frequencies transmitted from the orbiting ACES oscillator. Additionally, these methods for orbit determination ideally need to be sufficiently accurate to test General Relativity with the ACES clock ensemble.

Using GPS + Doppler data: Analysis of sample GPS data alone from receivers on board the ISS indicates that the position and velocity information are not sufficiently precise to meet the goal of testing predictions of General Relativity. However, in the U.S., accurate two-way ranging from NIST and Jet Propulsion Laboratory (JPL) can occur concurrently for short periods of time, several times each day. Combining the concurrent ranging with the GPS data should provide opportunities for very accurate position and velocity determinations. Additionally, many modern GPS receivers provide the Doppler frequency shifts of their transmission frequencies as well as pseudo-range and carrier phase information. We expect Doppler information also to be available from on-board GPS receivers during the ACES

	mission. To best take advantage of this set of ISS position/velocity information inputs, we developed theoretical expressions for two-way ranging and Doppler measurements, adapted for the specific NIST-ISS-JPL ranging geometries, and are working to implement the theory with specialized data filters handling the input data. Initial results are encouraging as we are now able to compute the velocity components of a given GPS receiver's antenna and preliminary analyses suggest that position determination should be possible to a few cm. Another approach to orbit determination would be to use the frequency data themselves. The ACES microwave links have been described in published literature; the scheme is based on R. F. C. Vessot's original Doppler cancellation method developed for a stable frequency transmitted between a launched rocket and Earth. With this method we can use differences in the transmitted frequencies measured at the respective receivers to yield information concerning the relative motion and gravitational potentials of the respective sources. We are presently considering incorporation of such frequency measurements into the software for orbit determination of the ISS. Testing the algorithms with GPS data sets and mock ACES sets: A powerful technique to test these algorithms is to determine the ISS orbit from real data sets. First we used GPS data sets supplied by the Johnson Space Flight Center to learn (i) how well we could really determine the ISS orbit and (ii) how to best handle the inevitable gaps in the data sets. In late November the Johnson Space Flight Center sent us approximately 48 hours of GPS position and velocity data taken by an ISS receiver. Indeed, we found large data gaps in the record, and also verified, as we had seen with an early data record provided by the same source, that the orbit of the ISS deviates from a simple Keplerian orbit model. To handle the gaps, we modelled the quadrupole potential of the Earth as an external perturbation. We found that the uncertainty in the radius of the orbit could be as high as a few hundreds of meters when there was a large gap in the data (10s of minutes). With a gap of few seconds between any two data points the error is less than a meter. Regarding the issue of the non-Keplerian orbit, we developed a framework to include external perturbations, in order to get an estimate of the orbital parameters as they change in time. The object is to make sure that it is possible to keep the uncertainties in velocity and position to less than 1 mm/s and 1 m respectively. We then tested our algorithms with mock data sets from the ACES team, with particular emphasis on data sets that correspond to the multiple passes when NIST and JPL are in common view (described above). This testing has helped us to modify the orbit determination codes that we've been working on here at NIST to match up to the inputs from the ACES data sets. While initial mock data sets were flawed, more recently supplied sets seem more realistic and we are presently evaluating them for optimization of our code/analysis techniques. [ED. NOTE (January 2016): Project continues with PI Chris Oates, Ph.D. See project with the same title with Oates as the PI for subsequent reporting.]

Bibliography Type:	Description: (Last Updated: 01/27/2016)
Articles in Other Journals or Periodicals	Oates CW, Ludlow AD. "Optical lattice clocks." Optics and Photonics News. 2015 Jan:26:36-43. , Jan-2015
Articles in Peer-reviewed Journals	Ludlow AD, Boyd MM, Ye J, Peik E, Schmidt PO. "Optical atomic clocks." Reviews of Modern Physics. 2015 Apr-Jun;87(2):637-701. http://dx.doi.org/10.1103/RevModPhys.87.637 , Jun-2015
Articles in Peer-reviewed Journals	Ludlow AD, Ye J. "Progress on the optical lattice clock." Comptes Rendus Physique. 2015 Jun;16(5):499-505. http://dx.doi.org/10.1016/j.crhy.2015.03.008 , Jun-2015
Articles in Peer-reviewed Journals	Heavner T, Donley E, Levi F, Costanzo G, Parker T, Shirley J, Ashby N, Barlow S, Jefferts S. "First accuracy evaluation of NIST-F2." Metrologia. 2014 Jun;51(3):174-82. http://iopscience.iop.org/article/10.1088/0026-1394/51/3/174 ; accessed 1/27/16. , Jun-2014
Articles in Peer-reviewed Journals	Ashby N, Barlow S, Heavner TP, Jefferts SR. "Frequency shifts in NIST Cs primary frequency standards due to transverse RF field gradients." Physical Review A. 2015 Mar;91(3):033624. http://dx.doi.org/10.1103/PhysRevA.91.033624 , Mar-2015